

Digital Integrated Circuits

Lab 3 Logical Effort

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Outline

- Lab Contents
- Report Requirements

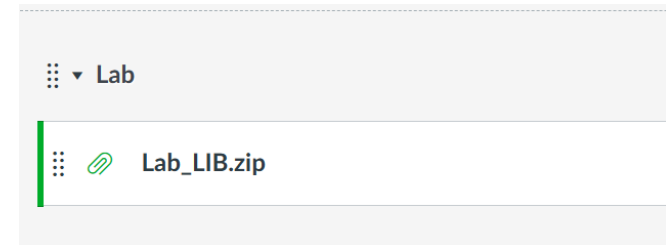
Outline

- ▣ Lab Contents
- ▣ Report Requirements

Device Models for Logical Effort Optimizaiton

❑ Use the Predictive Technology Model (PTM) to evaluate the DC characteristics of ptm16hp FinFETs

- PTM model is released in Lab section
- Use the high-power (HP) models
- For both n-channel and p-channel devices



Lab_LIB > FINFET > modelfiles > hp

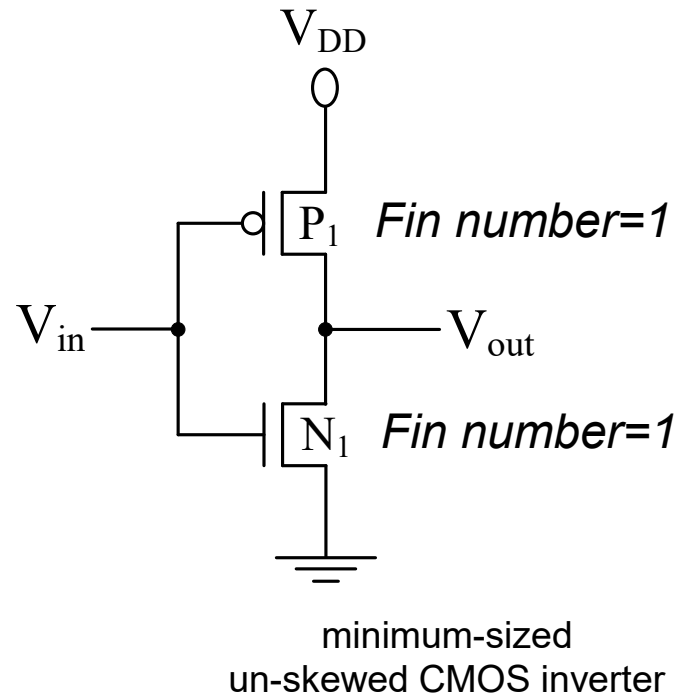
 16nfet.pm	2012/5/31 15:14	Perl Module 源文件	11 KB
 16pfet.pm	2012/5/31 15:14	Perl Module 源文件	11 KB
 20nfet.pm	2012/5/31 15:14	Perl Module 源文件	11 KB
 20pfet.pm	2012/5/31 15:14	Perl Module 源文件	11 KB

FinFET

Simulation Settings

- ❑ $L_g = 20\text{nm}$ for ptm16hp FinFETs
- ❑ Supply voltage is 0.75V for FinFETs
- ❑ Simulation temperature
 - $T = 25^\circ\text{C}$

Minimum sized INV for FinFET



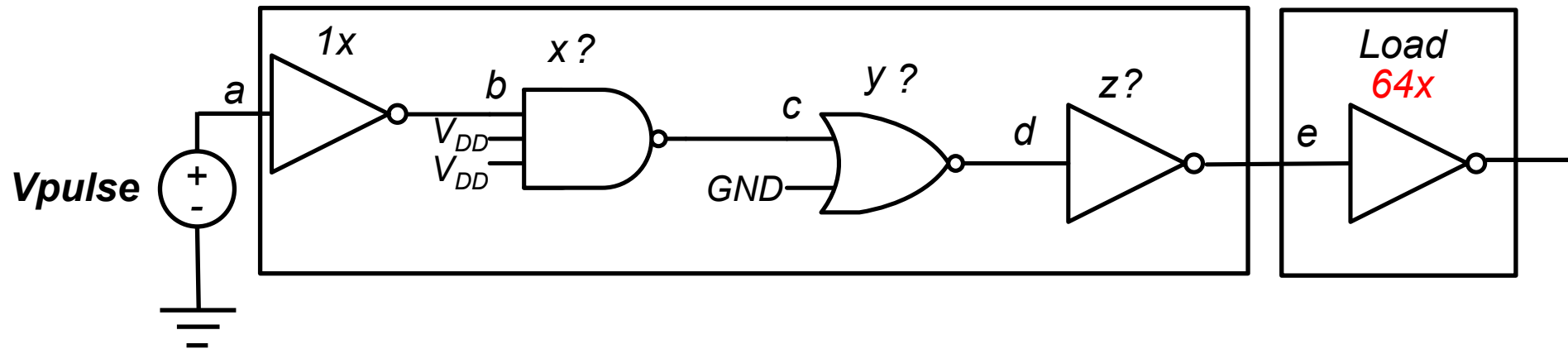
Minimum Sized INV	PMOS	NMOS
Fin Number	1	1
L	20n	20n
The fin number to ensure balanced driving current in pull-up and pull-down network		

Task 1: Use Logical Effort to Size Gates

- Size each transistor in the circuit below properly to optimize the speed **with FinFETs**, and construct the circuit with HSPICE

Vpulse	Value
V1	0V
V2	0.75V
Delay Time	900ps
Raise Time	100ps
Fall Time	100ps
Pulse Width	900ps
Period	2ns

Size the logic gates to minimize the propagation delay (a to e)



Hint: You can use the .MEASURE command to measure the propagation delay
For example (measure the propagation delay from a to e for L->H transition):

```
.MEASURE TRAN tpLH  
+ TRIG V(a)='0.5*SUPPLY' RISE=2  
+ TARG V(e)='0.5*SUPPLY' RISE=2
```

$$t_p = (t_{pHL} + t_{pLH})/2$$

Review: Logical Effort

- Concept of logical effort:

g – logical effort (逻辑努力): ratio of the input capacitance of the gate to the input capacitance of an inverter that can deliver the same output current, for a given load.

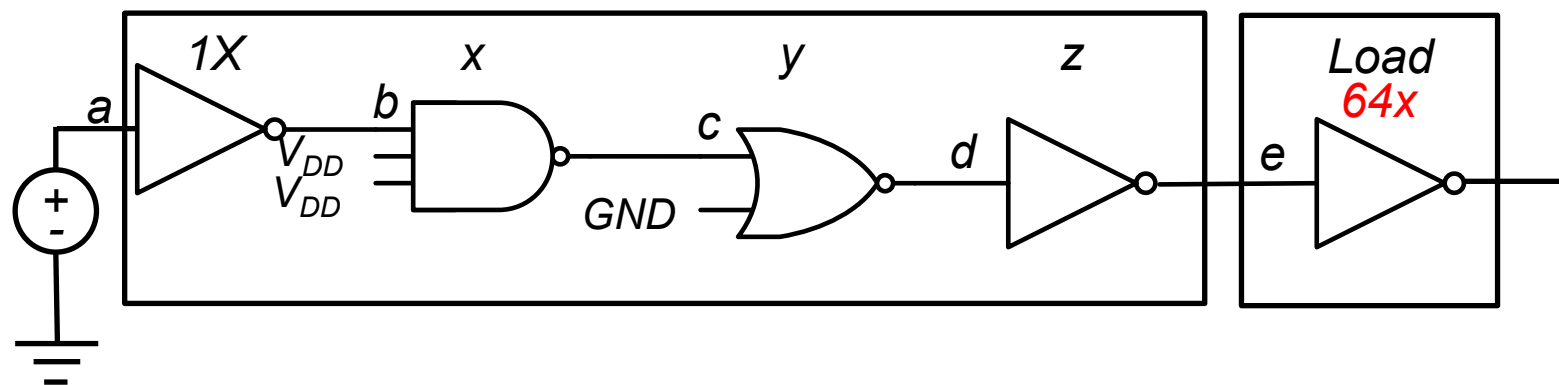
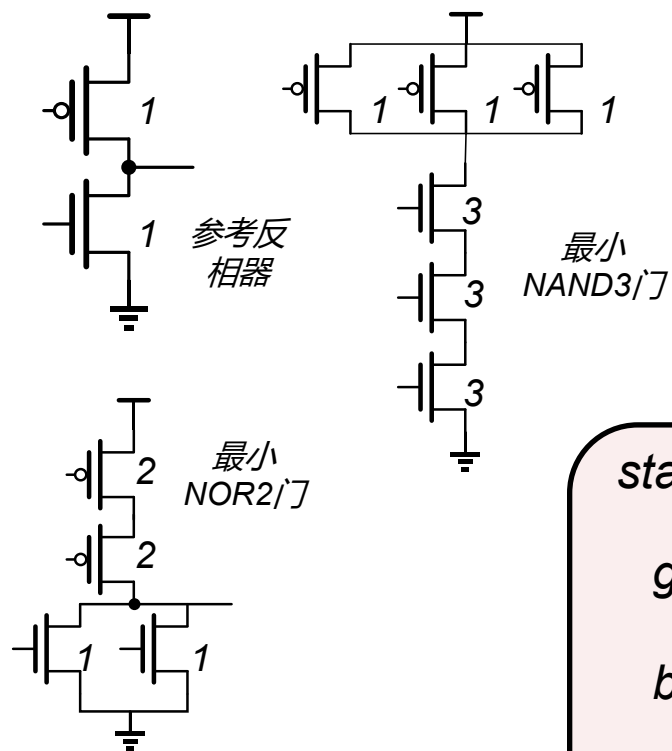
- Method of logical effort:

- Compute the path effort: $H = GFB$
- Compute the stage effort $h = H^{1/N}$
- Sketch the path with this number of stages
- Work from either end, find sizes:

$$C_{in} = C_{out} * g / h ;$$

□ 步骤提示:

- **步骤一:** 根据负载电容 C_{Load} 和逻辑门 (INV, NAND3, NOR2) 的逻辑努力计算理论每级最优的门努力 $h_{opt} = H^{1/4} = (GBF)^{1/4}$, 其中 $G = \prod_{i=1}^4 g_i$, $B = \prod_{i=1}^4 b_i$, $F = C_{load}/C_{in}$



stage	1	2	3	4		
g	$g_1=1$	$g_2=?$	$g_3=?$	$g_4=1$	$\Longrightarrow$	$G=?$
b	$b_1=1$	$b_2=1$	$b_3=1$	$b_4=1$	$\Longrightarrow$	$B=?$
f	$f_1=x/1$	$f_2=y/x$	$f_3=z/y$	$f_4=64/z$	$\Longrightarrow$	$F=64$

□ 步骤提示:

- **步骤二:** 根据步骤一得到的最优门努力 h_{opt} 计算每一级逻辑门(INV, NAND3, NOR2)的等效扇出 f_i (该门的外部负载和输入电容之比), 其中每级逻辑门的门努力 $h_i = g_i f_i b_i$, 则 $f_i = h_i / g_i b_i$

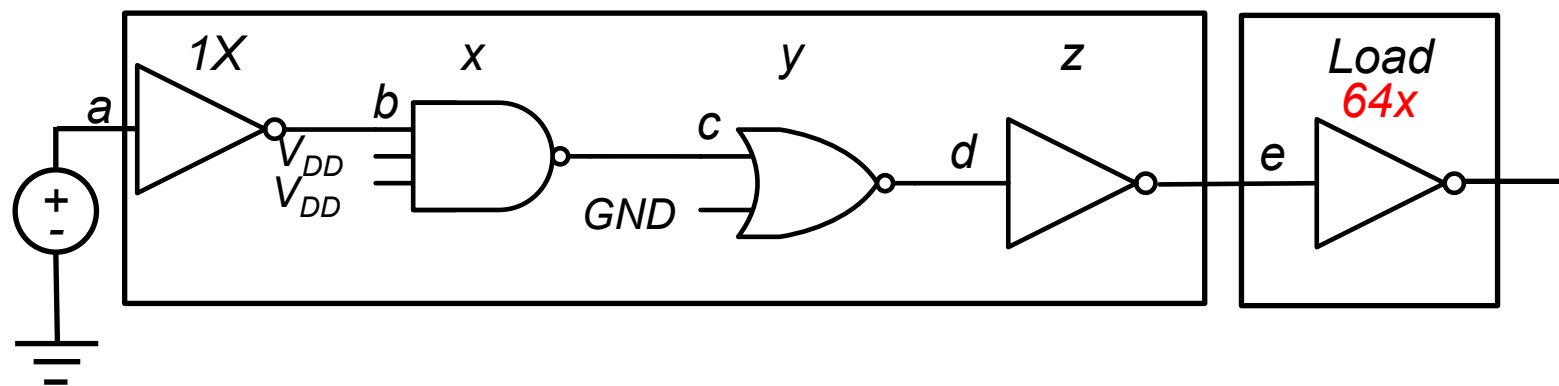
$$h_i = g_i f_i b_i$$

$$h_{opt} = H^{1/4} = (GBF)^{1/4}$$



$$f_1 = \frac{h_{opt}}{g_1 b_1} = x; \quad f_2 = \frac{h_{opt}}{g_2 b_2} = y/x;$$

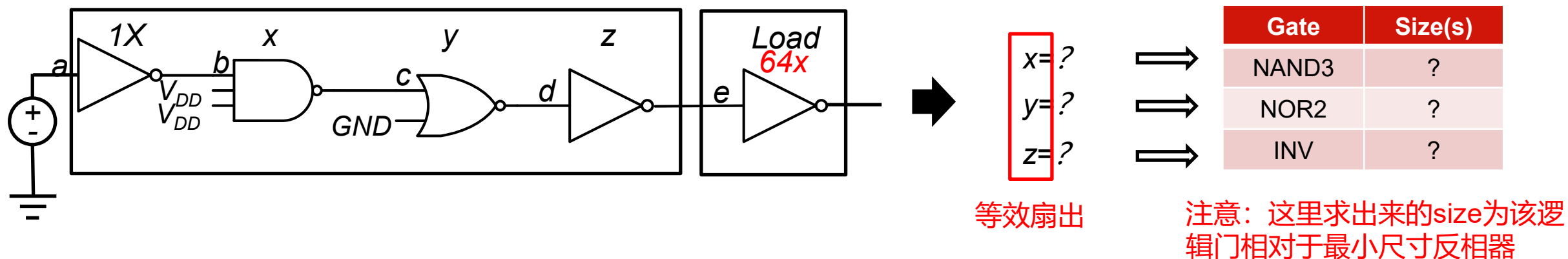
$$f_3 = \frac{h_{opt}}{g_3 b_3} = z/y; \quad f_4 = \frac{h_{opt}}{g_4 b_4} = 32/z;$$



stage	1	2	3	4
g	$g_1=1$	$g_2=?$	$g_3=?$	$g_4=1$
b	$b_1=1$	$b_2=1$	$b_3=1$	$b_4=1$
f	$f_1=x/1$	$f_2=y/x$	$f_3=z/y$	$f_4=64/z$
	$x=?$	$y=?$	$z=?$	

□ 步骤提示:

- **步骤三:** 根据步骤二计算得到的每级逻辑门的等效扇出 f_i , 计算链路上每一个逻辑门中晶体管的尺寸



- **步骤四:** 考虑到FinFET工艺下NMOS和PMOS晶体管的尺寸即fin number只能取整数, 同时保持上拉和下拉网络等效驱动强度相同, 进一步计算每级逻辑门 (INV, NAND3, NOR2) 的尺寸 s_i 取整的理论值, 并且可以考虑在理论值附近扫描

语法提示：Task1中逻辑门子电路定义

- Task1中每个逻辑门子电路定义时可以设置子电路中用到的参数值，如nfin

```
.subckt/.macro subnam n1 < n2 n3 ...> < parnam=val ...>
```

```
element1
```

```
element2
```

```
...
```

```
.ends/.eom <subnam>
```

- 子电路调用时，可以修改参数覆盖在子电路定义中分配的默认参数值，但会被.param 语句中设置的值覆盖，以便扫描不同的尺寸

Example:

***针对Task1，定义了一个反相器的子电路，具有 in 和 out 两个节点，并以 SIZE 作为参数，缺省值为 1**

```
.subckt INV in out SIZE=1
```

```
XPFET out in vdd vdd PFET L=20n nfin=SIZE
```

```
XNFET out in gnd gnd NFET L=20n nfin=SIZE
```

```
.ends
```

语法提示： Task1测量延迟时瞬态分析参数扫描方法

□ 单点分析

```
.TRAN var1 START=start1 STOP=stop1  
STEP=incr1
```

□ 数据驱动扫描

```
.TRAN DATA=devinf <SWEEP var2 start2  
stop2 incr2>
```

variable name (可以自己定义)

```
.DATA devinf width length thresh cap  
+ 50u 30u 1.2v 1.2pf  
+ 25u 15u 1.0v 0.8pf  
+ 5u 2u 0.7v 0.6pf  
+ ... ..  
.ENDDATA
```

*.DATA 语句提供了一种以具有相同行的
列并排排列的数据排列形式。*

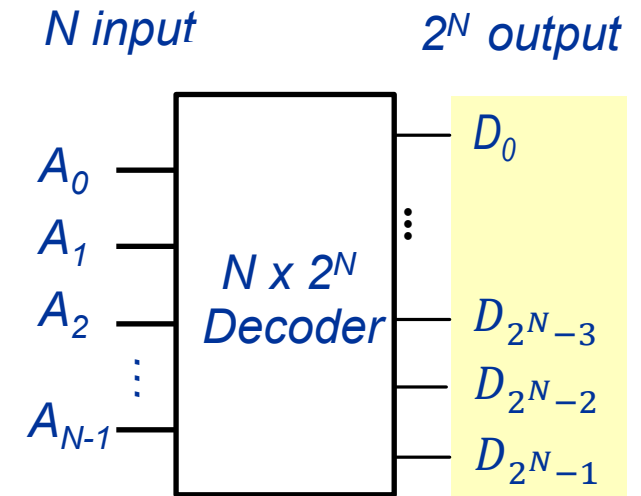
□ 变参数扫描：

```
.alter  
.param var=val
```

Task 2: Use Logical Effort to Optimize 4x16 Decoder

□ Background of decoder

- Logic function: **convert N-bit coded input to 2^N unique outputs**
- Can be used as address decoder/buffer chain in memory
 - Address line (input) has large fanout
 - Decoders typically have high electrical effort and branching effort
 - Decoder may have many stages, so the fastest design is the one that minimizes the logical effort



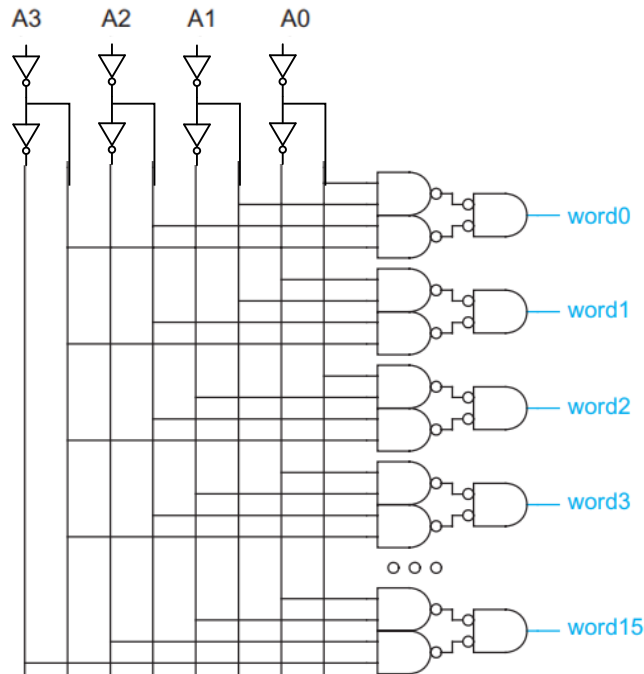
- With N input $A_0, A_1 \dots A_{N-1}$, output D_i :

$$\begin{aligned} D_0 &= \overline{A_{N-1}} \dots \overline{A_3} \overline{A_2} \overline{A_1} \overline{A_0} \\ D_1 &= \overline{A_{N-1}} \dots \overline{A_3} \overline{A_2} \overline{A_1} A_0 \\ &\dots \\ D_{2^N-1} &= A_{N-1} \dots A_3 A_2 A_1 A_0 \end{aligned}$$

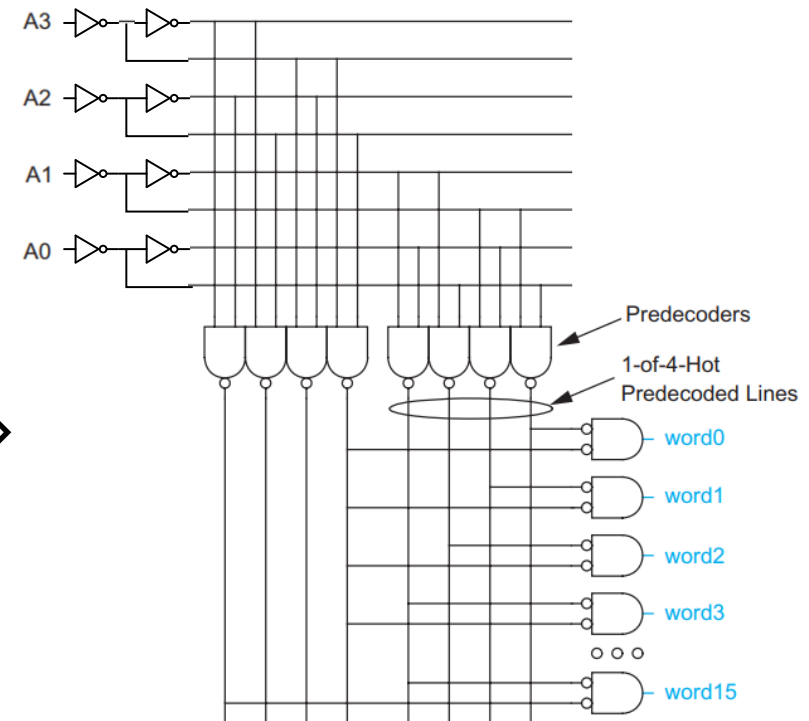
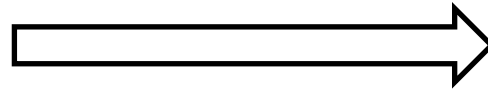
The complement signal can be obtained by inverter

Task 2: Use Logical Effort to Optimize 4x16 Decoder (Cont.)

- An example of 4x16 decoder design:



Factoring the common
2-input NANDs out



Precoding technique
to reduce the area

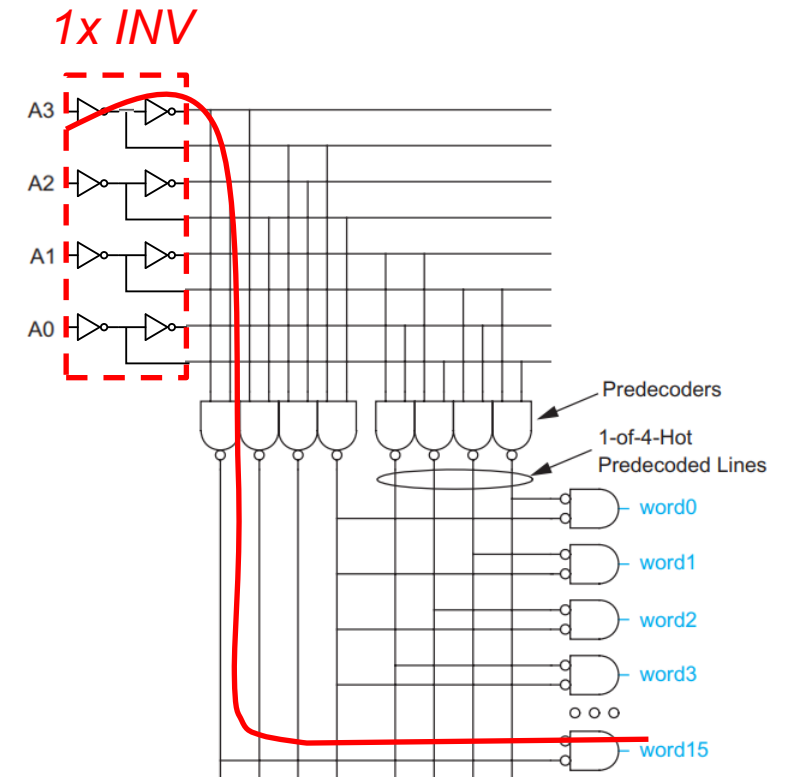
4-input AND function is built from a pair of
2-input NANDs followed by a 2-input NOR

But, many NAND gates share the same
inputs and are redundant

Task 2: Use Logical Effort to Optimize 4x16 Decoder (Cont.)

□ For a 4x16 decoder, optimize the critical path to minimize the propagation delay of decoder

- Use the logical effort method
- Assume that the input inverter/buffer is minimum sized
- Assume that each output of decoder has a load of 128X minimum sized Finfet inverter
- You can **insert buffer** at the output if necessary



Review: Adding inverters to end of path can give minimum delay

- 通过一条路径末端添加偶数个反相器，既不会改变功能与极性，而且还可能使得组合逻辑门链路的总延时最小

新路径的延时:

$$D = NH^{\frac{1}{N}} + \sum_{i=1}^{n_1} p_i + (N - n_1)$$

对N求导并使导数为0, 求得最优级数:

$$\frac{\partial D}{\partial N} = -H^{\frac{1}{N}} \ln H^{\frac{1}{N}} + H^{\frac{1}{N}} + p_{inv}$$

→ $h(1 - \ln h) + p_{inv} = 0$, where $h = H^{\frac{1}{N}}$

插入的反相器
数量:

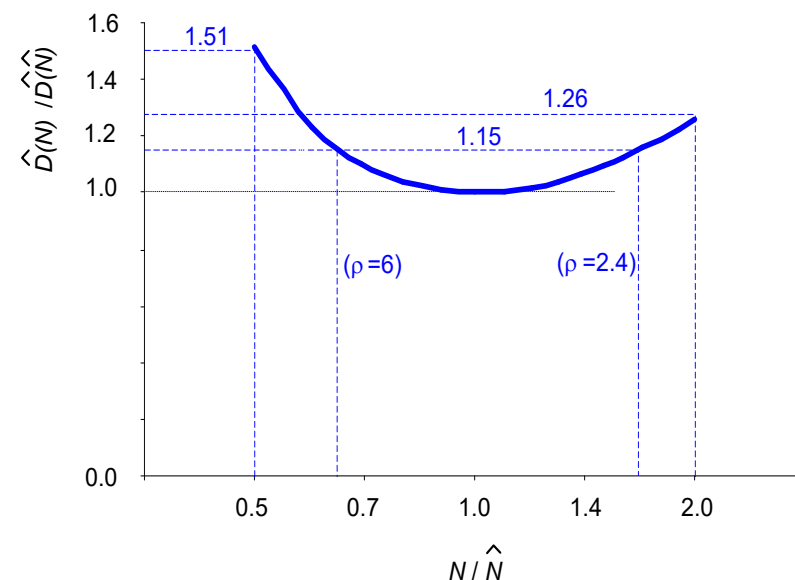
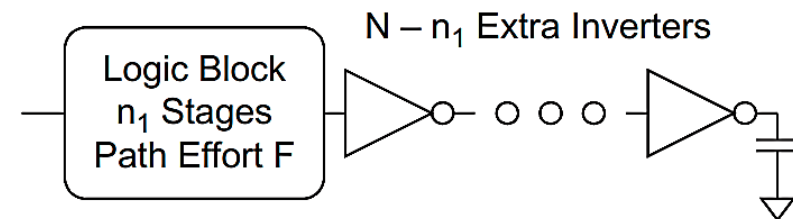
$$N = \log_h H$$

$$D = Nh + \sum_{i=1}^{n_1} p_i + (N - n_1)$$

$$= h \log_h H + \sum_{i=1}^{n_1} p_i + (\log_h H - n_1)$$

当 $p_{inv} = 1$ 时, 采用数值方法可求解得到:

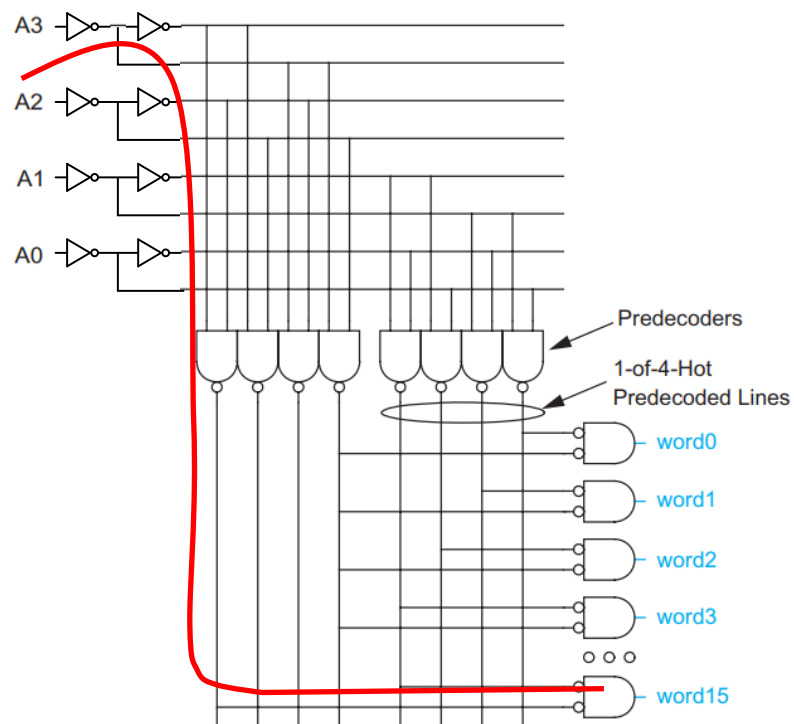
$$h = 3.59$$



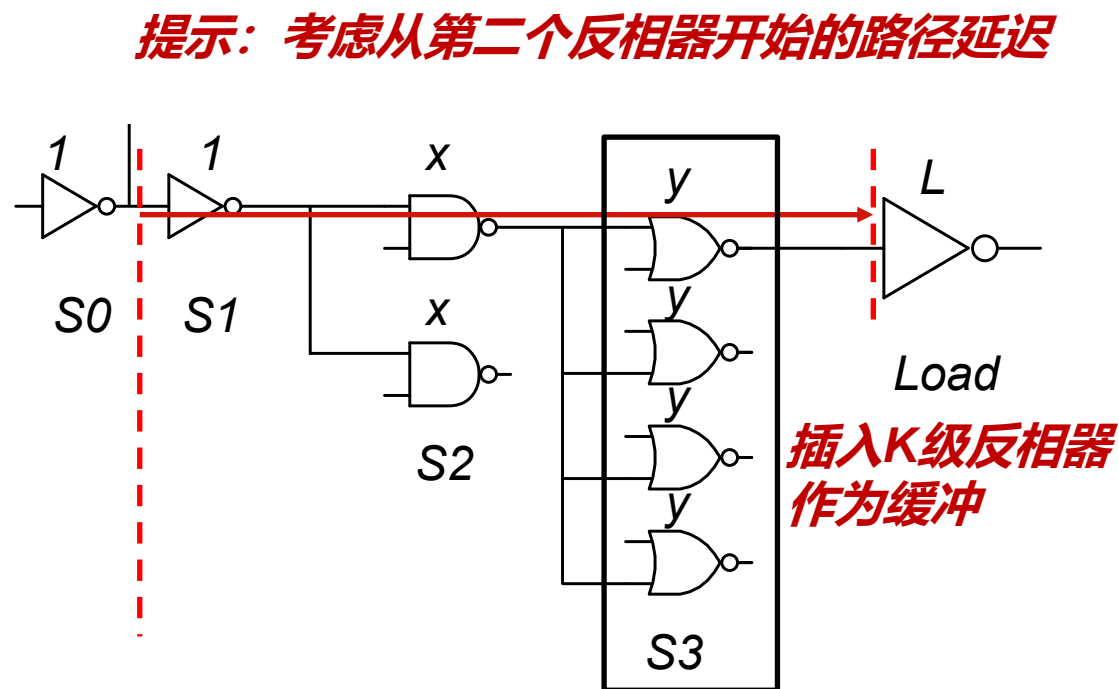
$2.4 < h < 6$ gives delay within 15% of optimal
We can be sloppy! I like $h = 4$

□ 步骤提示:

- **步骤一:** 根据解码器的布尔表达式分析得到组合逻辑电路的**关键路径** (最长路径分析), 如 $\text{word1} = \overline{A_3} \overline{A_2} \overline{A_1} A_0 = \overline{\overline{\overline{A_3}} \cdot \overline{\overline{A_2}} + \overline{\overline{A_1}} \cdot \overline{\overline{A_0}}}$, 针对于每个输入(如 A_0)信号需要经过两级反相器INV, 一级NAND2门和一级NOR2门(含有分支努力), 以及负载电容前的k个反相器构成的缓冲

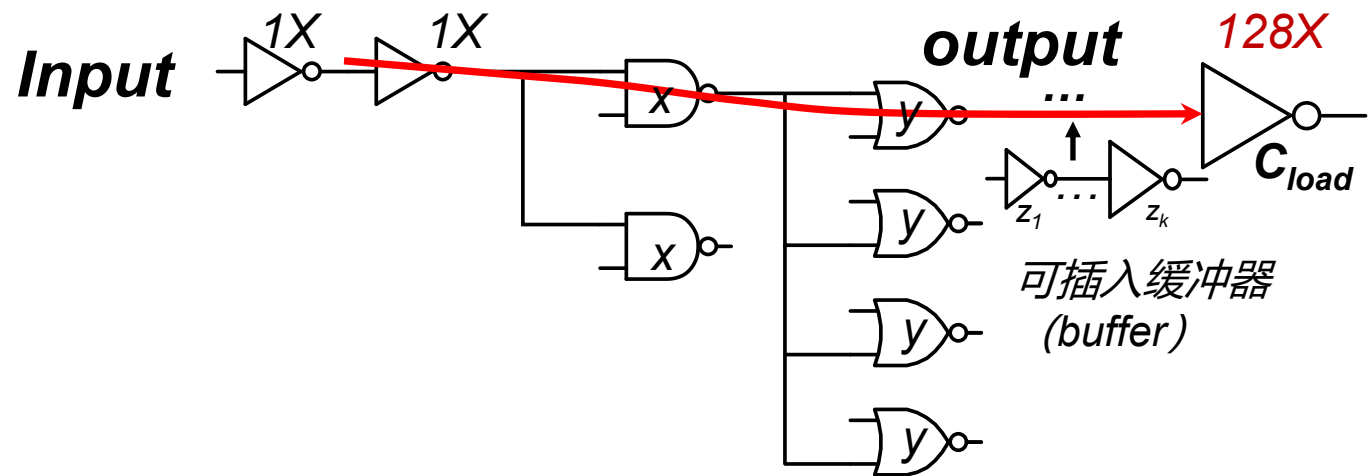


➡
关键路径分析



□ 步骤提示:

➤ 步骤二: 在步骤一分析得到关键路径的基础上假定每级逻辑门(INV, NAND2, NOR2)合适的门努力 h_{opt} , 并计算得到合适的逻辑门级数 N (即输出负载前插入多少反相器) $N=\log_{h_{opt}} H$



提示: 可假定最优的门努力 h_{opt} 为4或附近的整数, 计算最优级数 N

$h_i = g_i f_i b_i$
 $H = GBF$
 $G = ?$
 $B = ?$
 $F = ?$
 $H = ?$

stage	1	2	3	4	...	N
g	1	...	...	...	...	1
b	2	4	1	1	...	1
f	$x/1$	y/x	z_1/y	z_2/z_1	...	$64/z_k$
h	...	...	...			

$n_1=3$

□ 步骤提示:

- **步骤三:** 根据假设的最优门努力 h_{opt} 计算每一级逻辑门(INV, NAND2, NOR2)的等效扇出 f_i , 其中每级逻辑门的门努力 $h_i = g_i f_i b_i$, 则 $f_i = h_i / g_i b_i$, 再计算出链路上每一个逻辑门中晶体管的尺寸

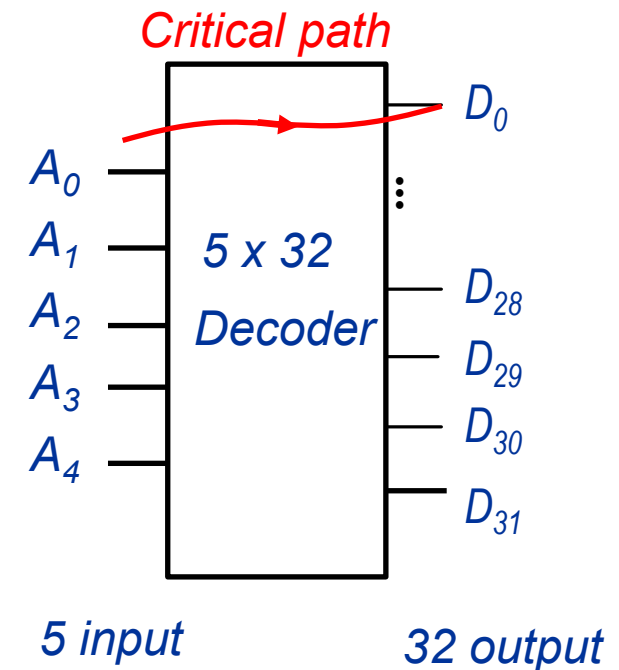
Stage1	Stage2	Stage3	Stage4	Stage5	Stagek
第二级反相器	NAND2	NOR2	增加第一级反相器	增加第二级反相器	...
1	?	?	?	?	...

- **步骤四:** 考虑到FinFET工艺下NMOS和PMOS晶体管的尺寸即fin number只能取整数, 同时保持上拉和下拉网络等效驱动强度相同, 进一步计算每级逻辑门(INV, NAND2, NOR2)的尺寸 s_i 取整的理论值, 并且可以考虑在理论值附近扫描

Task 3 (Optional) : Use Logical Effort to Optimize 5x32 Decoder

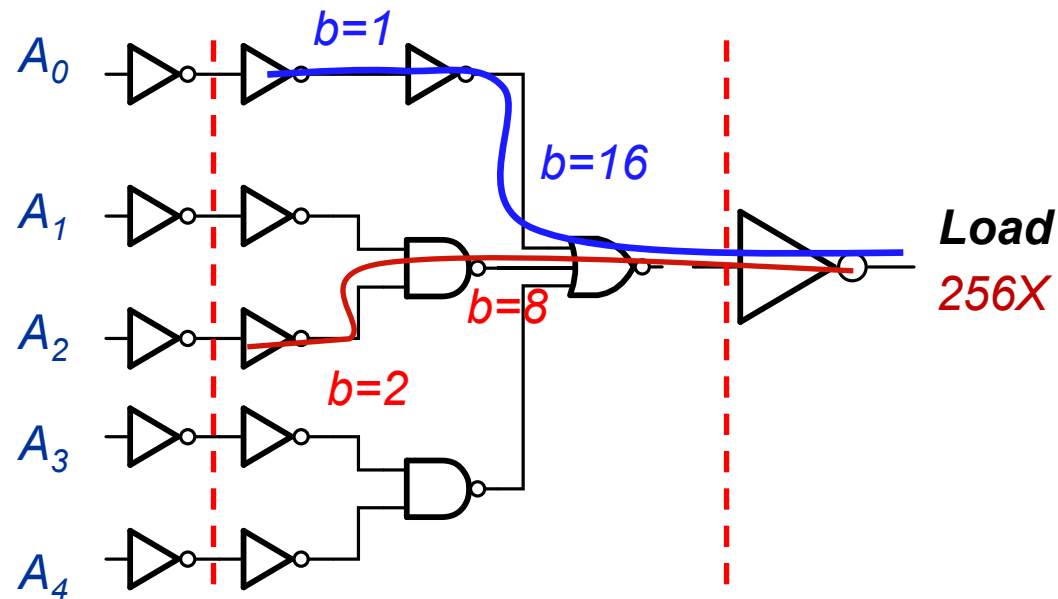
□ For a 5x32 decoder, optimize the critical path to minimize the propagation delay of decoder

- Use the logical effort method
- Assume that the input inverter/buffer is minimum sized
- Assume that each output of decoder has a load of 256X minimum sized Finfet inverter
- You can insert buffer at the output if necessary



□ 步骤提示:

➤ **步骤一:** 根据解码器的布尔表达式分析得到组合逻辑电路的关键路径（最长路径分析），如 $\text{word1} = \overline{A_4} \overline{A_3} \overline{A_2} \overline{A_1} A_0 = \overline{\overline{A_4} \cdot \overline{A_3}} + \overline{\overline{A_2} \cdot \overline{A_1}} + \overline{\overline{A_0}}$ ，因此需要对于每个输入(如 A_0)信号需要经过两级反相器INV，一级NAND2门和一级NOR3门(含有分支努力)，以及负载电容前的k个反相器构成的缓冲



可以看到一共有两种不同支路，对于这两种不同支路分别计算出H1和H2，再计算出对应的最优优化级数为N1和N2

S1	S2	S3	S4	S5
1x	?	?	?	?

S6	S7
?	?

➤ **步骤二~步骤四:** 与Task2的分析类似

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- Report Requirements

Report Requirement

- ❑ Write your lab report like writing a technical document (readable, comprehensive analysis, no typo...)
- ❑ You may include
 - Lab procedures (e.g., screenshots of the critical steps)
 - Lab results (e.g., schematic view, symbol of inverter, screenshots of the simulation waveforms)
 - Observations and conclusions

Submission

- ❑ You need to submit your report and code
 - Name of report (in PDF format):
lab3_report_[Name]_[Student No.].pdf
 - Name of code (compressing the files):
lab3_code__[Name]_[Student No.].zip
- ❑ Please upload your report to Canvas course website
- ❑ Submission of lab1 report and code will be due on
12.14-2025